

Lesson 4 ROS2 Development Environment Configuration

Note: The provided Raspberry Pi image comes pre-installed and pre-configured with the development environment. This lesson is only provided for learning and reference purpose only.

During the subsequent learning and development processes, users can improve their efficiency by optimizing development environment and installing plugins.

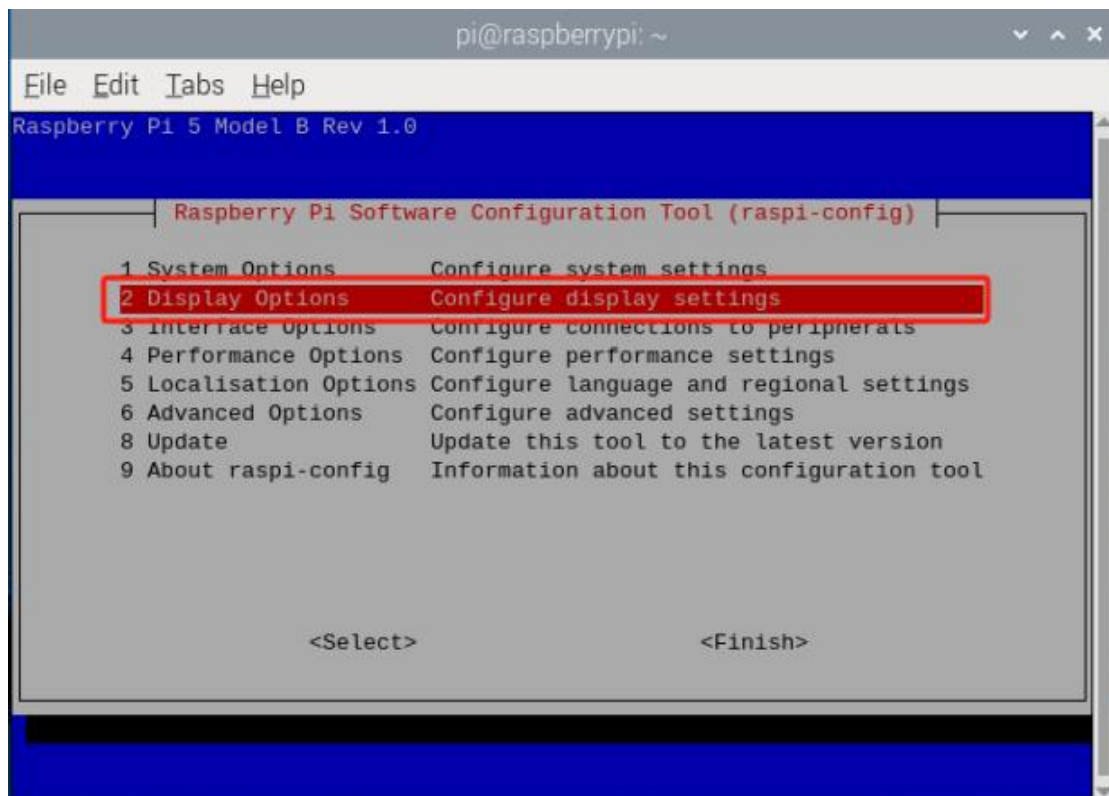
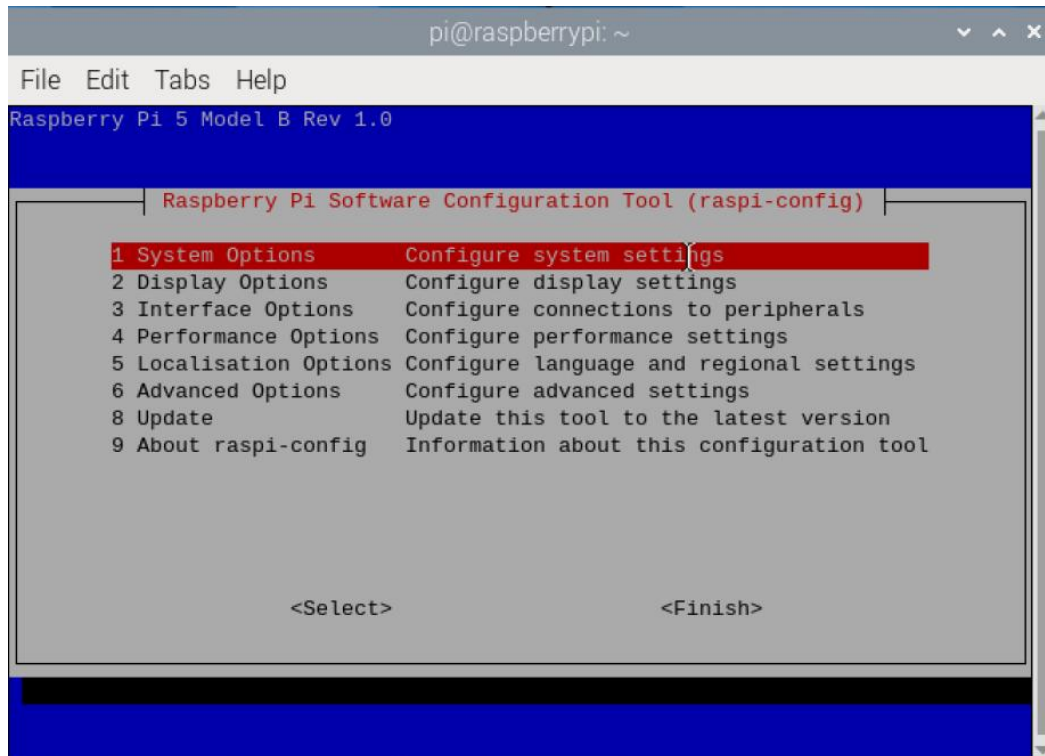
1. Adjust Resolution

- 1) Press “**Ctrl+Alt+T**” to open a command-line terminal and input the command “**sudo raspi-config**”, then press Enter.

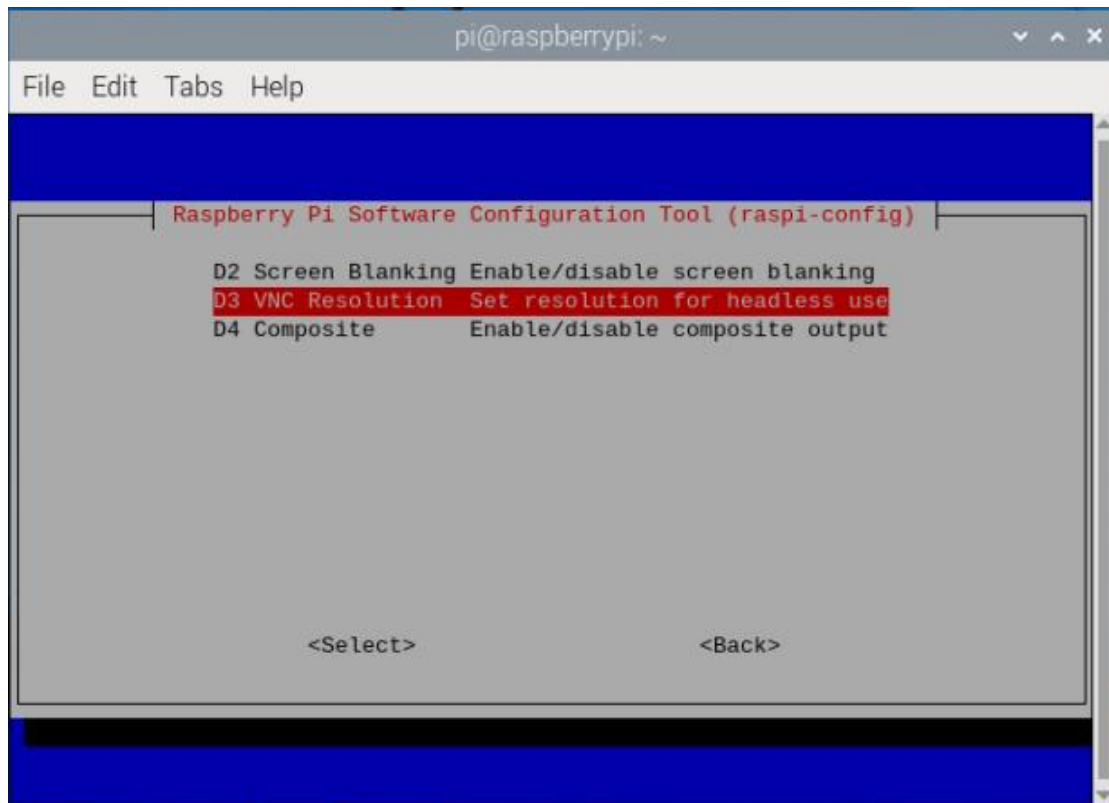


```
pi@raspberrypi: ~
File Edit Tabs Help
pi@raspberrypi:~$ sudo raspi-config
pi@raspberrypi:~$
```

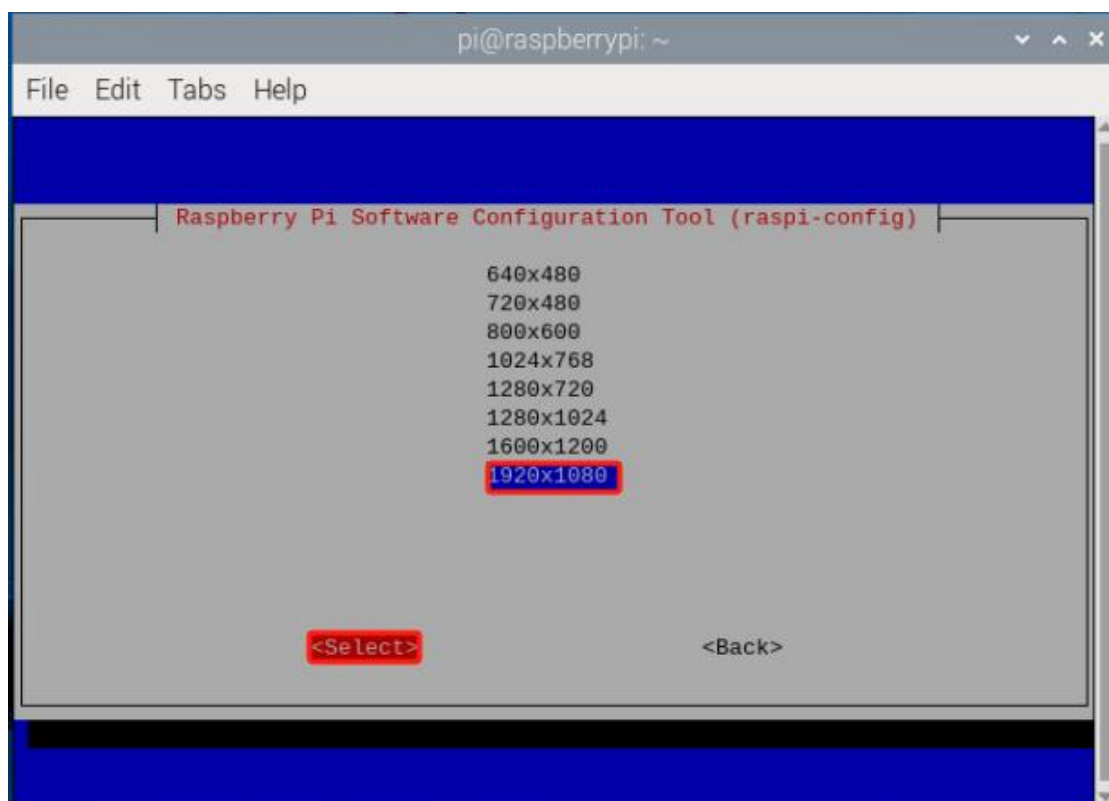
- 2) Select “2. Display Options” using “**↑ ↓**” keys and press “Enter” to confirm.
“Esc” key can be used to go back to the previous level.



3) Select “D3 VNC Resolution”, and then press “Enter” to confirm.



- 4) Select “1920*1080” and “<Select>” in sequence, then press “Enter” to confirm.

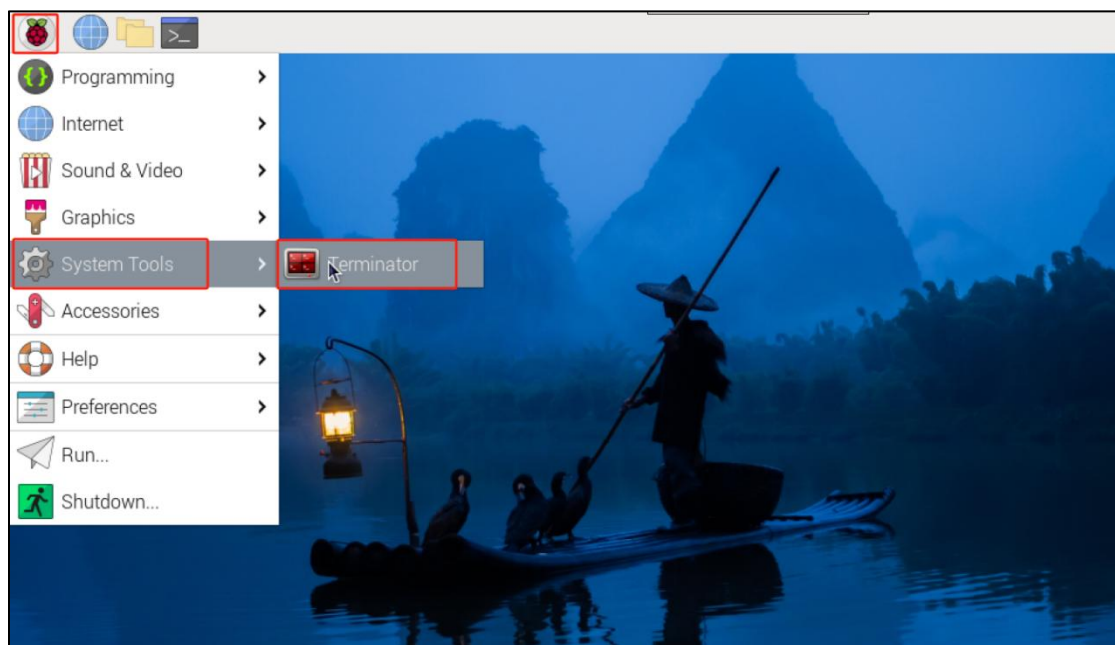


5) When the modification is completed, Raspberry will restart.

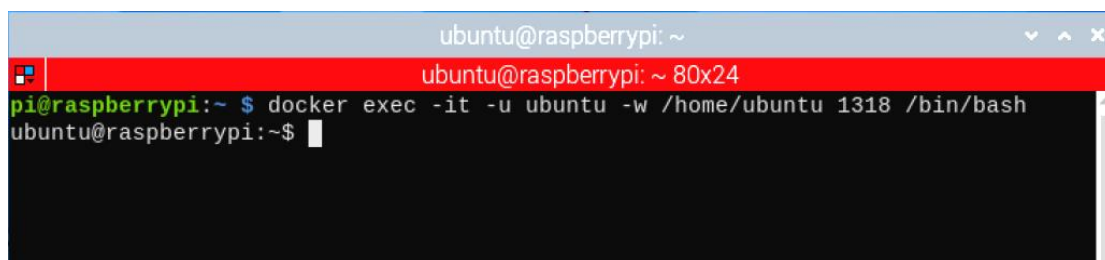
2. Tool Usage

During the subsequent ROS development, it is necessary to open multiple terminals. In this point, terminator tool will be utilized.

1) Click  in the upper-left corner, and select System Tools and Terminator in sequence.

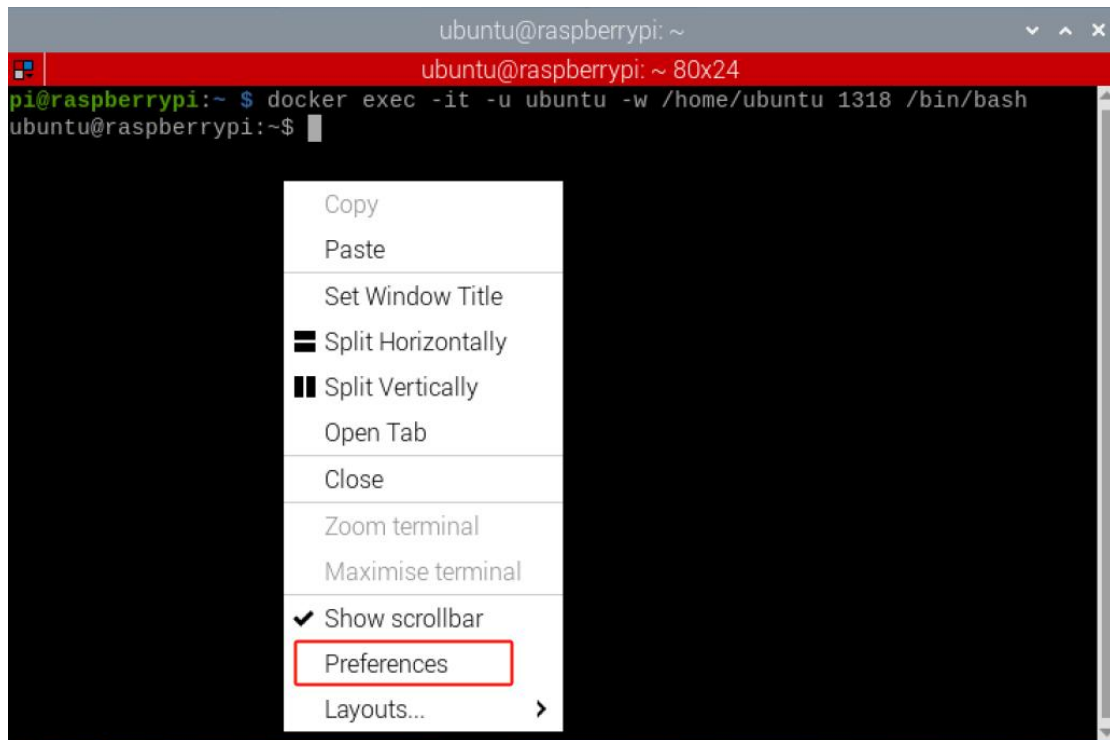


2) Enter “**docker exec -it -u ubuntu -w /home/ubuntu 1318 /bin/bash**” to enter the container.

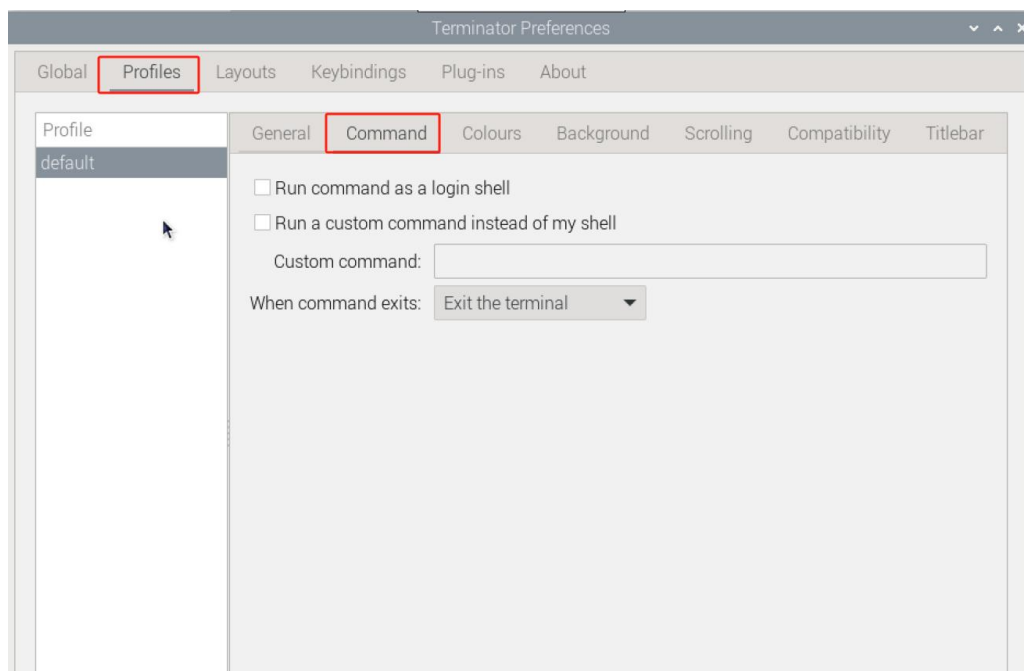


Every time before entering the container, it is inconvenient to enter command in the terminator terminal. You can set the command to enter the container in the terminator tool.

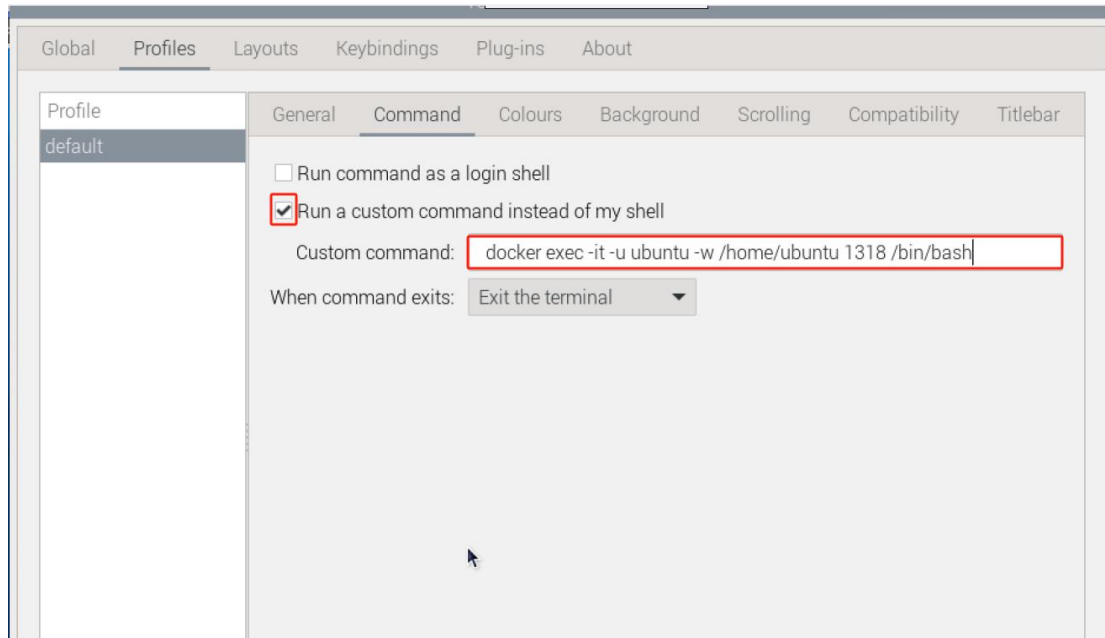
- 3) Right-click on the terminator window, and then select "Preference".



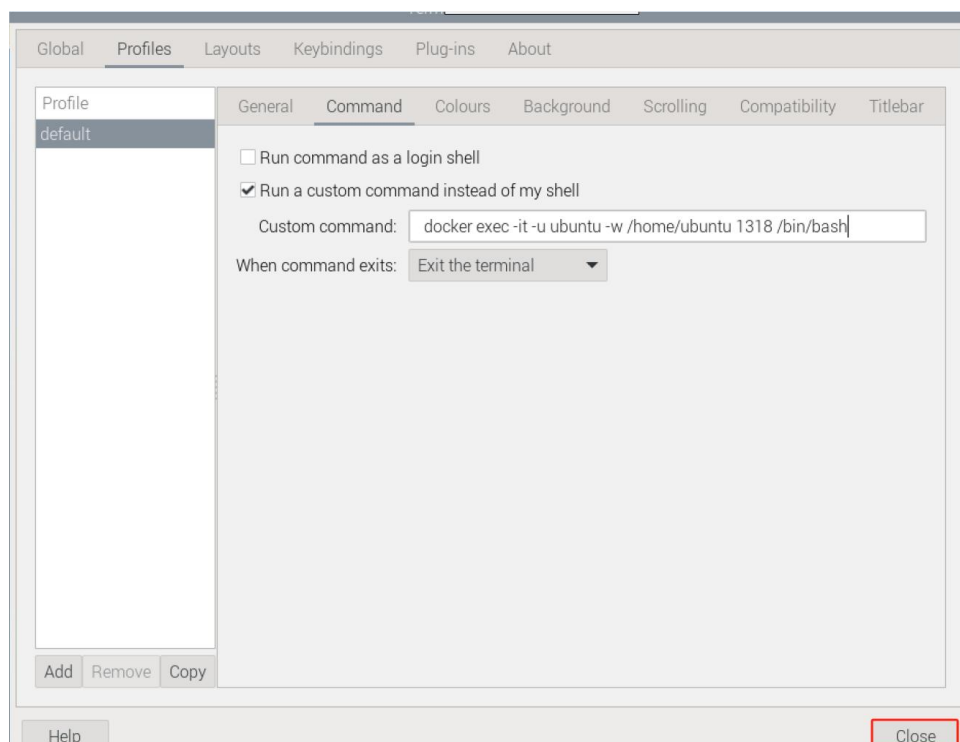
- 4) Select **Profiles**→**Command** in sequence.



5) Check the “Run a custom command instead of my default” option, and then enter the accessible command of container “**docker exec -it -u ubuntu -w /home/ubuntu 1318 /bin/bash**”. (Note: 1318 is the container ID containing ROS2 environment.)



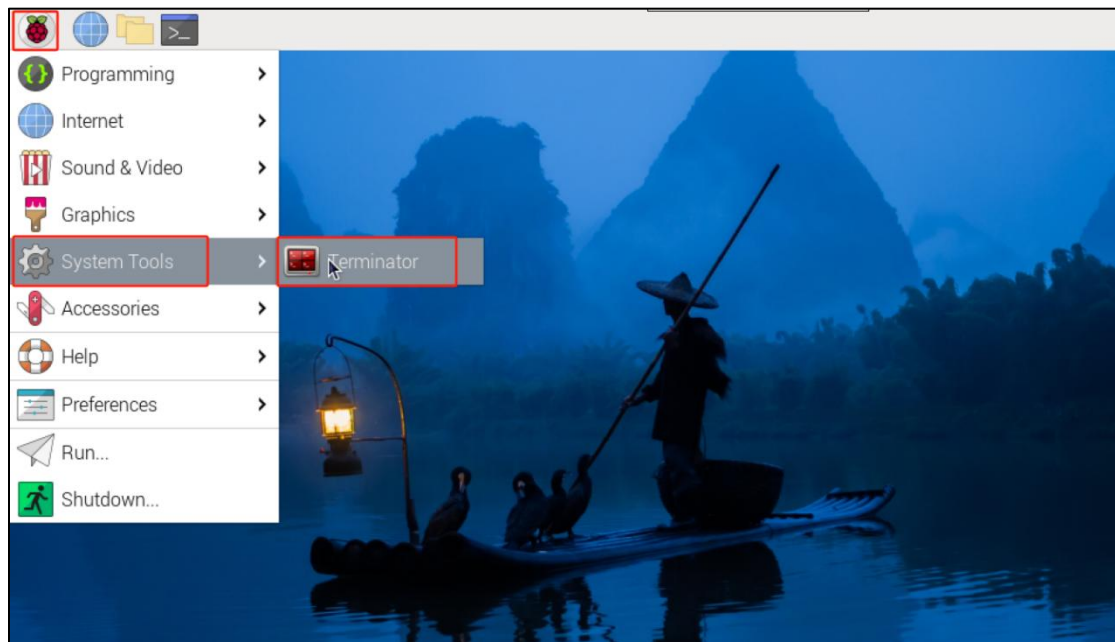
6) Click “Close”. This way, every time you open the terminal, you can directly enter the container with ROS2 environment.



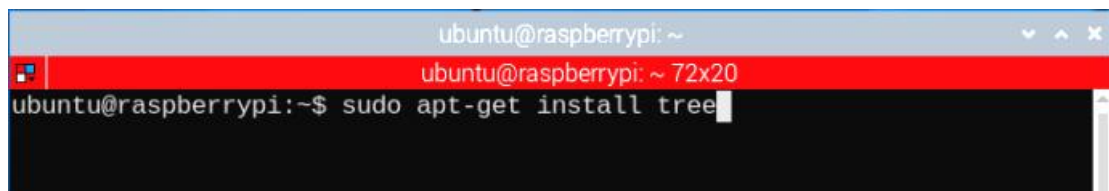
3. Tree Tool Installation

The Tree is a command-line tool, used to list the hierarchical structure of directories and files in a tree-like format. It helps users visualize the directory structure of the file system in a clear manner, including files and sub-directories.

- 1) Click  in the upper-left corner, select **System Tools** and **Terminator** in sequence.



- 2) Enter “**sudo apt-get install tree**” to install Tree tool.



- 3) After installation, enter “**tree -L 1**” to display the first-level directories structure of the current folder.


```
ubuntu@raspberrypi:~$ tree -L 1
.
├── build
├── core
├── frames_2024-01-29_06.52.19.gv
├── frames_2024-01-29_06.52.19.pdf
├── frames_2024-01-29_06.53.08.gv
├── frames_2024-01-29_06.53.08.pdf
├── frames_2024-01-29_08.47.41.gv
├── frames_2024-01-29_08.47.41.pdf
├── hiwonder_ws
├── install
└── log

4 directories, 7 files
```

4. pip Package Installation

Python-pip is a package for Python3's package management tool "pip". pip is the official package management tool for Python, used to install, upgrade and manage Python package .

Enter the command "**sudo apt-get install python3-pip**" to install pip.

```
ubuntu@raspberrypi:~$ sudo apt-get install python3-pip
```

5. transforms3d Library Installation

Transforms3d is a Python library used for 3D transformations and rotation matrices. It provides a set of functions and classes for performing various transformation operations in three-dimensional space, such as rotation, translation and scaling. The traforms3d library has a simple and user-friendly interface that can be used for handling 3D graphics, robot kinematics, computer vision, and other fields. It supports various common presentation of rotations, such as Euler angles, quaternions, and rotation matrices, and

provides conversion functions for converting between these representations.

Enter the command “**sudo pip3 install transforms3d**” to install Python library.

```
ubuntu@raspberrypi:~$ sudo pip3 install transforms3d
```

Note: Prior to the installation of transforms3d library, you need to first install pip package.

6. turtle-tf2-py& tf2-tools Libraries Installation

turtle-tf2-py and tf2-tools are two packages in ROS, used to perform coordinate transformation-related functionalities in Python using TF2 (Transform Library).

turtle-tf2-py provides a lightweight TF2 client library for Python, enabling easy coordinate transformation operations in ROS. By using ros-humble-turtle-tf2-py, you can listen for coordinate transformations, query coordinate transformation, perform coordinate transformations, and more.

tf2-tools provides some TF-related practical tools. It includes commonly used functionalities such as interpolation of coordinate transformations, publishing and listening to coordinate transformations, and visualization of coordinate frames.

Enter “**sudo apt install ros-humble-turtle-tf2-py ros-humble-tf2-tools**” to install turtle-tf2-py and tf2-tools libraries (humble is a version number for ROS2.)

```
ubuntu@raspberrypi:~$ sudo apt install ros-humble-turtle-tf2-py ros-humble-tf2-tools
```

7. Gazebo Installation

Gazebo is a powerful open-source 3D robot simulation software that helps us quickly create and test various real-world scenarios on a computer. It comes with a built-in precise physics engine to simulate the interactions of dynamic objects in the scene, such as gravitational forces and friction. It also provides excellent 3D visualization effects, allowing us to clearly see the simulation process through a visual client. Most importantly, Gazebo integrates seamlessly with ROS, allowing us to develop and test ROS nodes directly within it. This enables us to easily transfer simulation code to physical robots.

Enter “**sudo apt-get install ros-humble-ros-gz**” to install Gazebo (humble is a ROS number for ROS2.)

```
ubuntu@raspberrypi:~$ sudo apt-get install ros-humble-ros-gz
```